INCH-POUND
A-A-50717
15 March 89

COMMERCIAL ITEM DESCRIPTION

ATTENUATOR VARIABLE
DC TO 18 GHz

The General Services Administration has authorized the use of this Commercial Item Description

Scope This commercial item description (CID) describes a step attenuator. This CID is meant as a minimum requirement for the attenuator variable in which only those manufacturers that meet or surpass the following requirements are qualified this CID

Classification Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. Equipment shall be type III, class 5, style E and color R in accordance with the latest MIL-T-28800. This CID makes provision for the bid sample testing in accordance with the latest MIL-T-28800

Applicable documents

Specifications and standards The following specifications, standards and handbooks form a part of this CID to the extend specified herein. Unless otherwise specified the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto cited in the solicitation

SPECIFICATIONS

MILITARY

MIL-T-28800 Test Equipment for use with Electrical and Electronic Equipment, General Specification for

(Copies of specifications, standards, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Other publications The following document forms a part of this CID to the extent specified herein. Unless otherwise specified herein. Unless otherwise specified the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation

FSC 5985

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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS

IEEE STD 587-1980

(Application for copies of IEEE publications should be addressed to the Institute of Electrical and Electronic Engineers, 345 East 47th Street, New York, NY 10017).

Order of precedence. In the event of conflict between the text of this CID and the references cited herein (except for the associated detail specifications, specification sheets for MS standards) the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Requirement.

Salient characteristics. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

1. Safety. The equipment shall be UL (Underwriters Labortories) listed and approved and/or shall comply with the type III safety requirements of MIL-T-28800.

a. Leakage Current Protection. Protection shall be provided from leakage current in excess 3.5 milliamperes (mA) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es).

b. Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated for at least four (4) times the operating voltage.

c. Type III Ventilation openings.

(1) Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts.

(2) Top ventilation openings shall incorporate small mesh screening or similar design to prevent the expulsion of hot electronic particles due to failure of electronic devices such as capacitors.

2. Parts, materials, and processes. Except for marking, the step attenuator shall comply with the type III parts, materials, and process requirements of MIL-T-28800. Nonstandard parts approval requests shall be in accordance with MIL-T-28800. Front panel marking shall be in accordance with MIL-T-28800.

a. Restricted materials. The equipment shall comply with the restricted material requirements of MIL-T-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

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3 Design and construction Unless otherwise specified herein, the equipment shall comply with the type III design and construction requirements of MIL-T-28800 The equipment shall be of solid-state construction Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuites where construction is susceptible to leakage or corona, indicators and associated controls shall be in close proximity to each other, all potentiometer controls shall have at least a 270 degree rotation range (4 71 radians)

a Type of construction Construction of the step attentuator shall be of a nonolithic or a seperate component design

b Signal Input and Output Connectors Unless otherwise specified all signal input and output connectors shall be type N female

c Internal Cooling The equipment shall be capable of operating continuously for twenty-four (24) hours at 55°C If forces air is used it shall be as specified in MIL-T-28800

4 Electrical Power Sources The equipment shall operate from nominal commercial, military, and shipboard power sources of 100 110, 120 200 220 and 240 volts ac (±10 %) at line frequencies of 50 Hz to 400 Hz

a Electrical Power Connectors The equipment shall operate with a ground at the primary source or either side of the single phase input

b Power Consumption Maximum power consumption of all components combined shall not exceed 2 watts average each or 50 watts peak without damage or change in accuracy

c Fuses Fuse(s) shall be provided on the exterior rear panel for replacement without entering the enclosure The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements

d Surge and Transient Voltage Protection The Step Attenuator shall contain surge and transient voltage protection internal of the enclosure The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per type III solid state tolerance band of MIL-T-28800 the surge voltage protection circuit shall conform to IEEE Std 587-1980 The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection Normal operation of the Step Attenuator may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 percent and frequency transient conditions greater than ±5 percent from within the nominal type III steady-state tolerance bands of MIL-T-28800 damage (other than fuse failure) to the Step Attenuator shall not occur during voltage transient conditions greater than ±20 percent and frequency transient conditions greater than ±5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) specified in MIL-T-28800 Normal operation of the Step Attenuator shall be automatically resumed when the transient has passed and any blown fuses

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are replaced. Blown fuses shall be externally accessible. Testing shall be in accordance with MIL-T-28800.

5. Dimensions and Weight. The size and weight of individual component shall be consistent with current commercial capabilities, refer to paragraphs 5.a and 5.b of this CID.

a. Equipment dimensions. The nominal dimensions per component for the step attenuator should be approximately 4.75 inches height x 4.60 inches wide x 5.88 inches deep.

b. Equipment weight. The weight including all accessories excluding manual) per component, shall not exceed 4.5 lb.

6. Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the style E enclosure requirements of MIL-T-28800.

7. Marking and identification.

a. Marking. The visible surface adjacent to panel facilities such as variable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation hereof, indicating the purpose or use of the part.

(1) Panel marking. The front panel marking shall comply with the type III parts, materials, and processes panel marking requirements of MIL-T-28800, however, marking identification processes other than the type III requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with this CID.

b. Items to be marked (if applicable).

(1) Volume and weight (including accessories) shall be marked on three external surfaces of the enclosure.

(2) The type designation of the AN nomenclature of the step attenuator shall be stenciled on the exterior or mainframe on at least three adjacent surfaces, on the transient case on two opposite sides. Marking shall be in block letters approximately 2 inches in height.

8. Environmental requirements.

a. Temperature requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class 5 temperature requirements specified by MIL-T-28800 and as listed below.

(1) Non-operating temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -10°C to +75°C.

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b. Humidity requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for class 5 equipment as specified by MIL-T-28800.

c. Vibration requirements. The equipment shall meet the class __ vibration requirements as specified by MIL-T-28800.

d. Mechanical shock requirements.

(1) Shock, functional. The equipment shall meet the class 5 shock, pulse requirement as specified by MIL-T-28800.

(2) Transit drop. The equipment shall meet transit drop requirement as specified by MIL-T-28800.

(3) Bench handling. The equipment shall meet the requirements as specified by MIL-T-28800.

e. Electromagnetic interface (EMI). The equipment shall comply with MIL-T-28800, Table XII.

f. Magnetic environment. The equipment shall comply with the magnetic environment requirements of MIL-T-28800.

9. Reliability. Reliability verification testing, shall be in accordance with the type III, class 5 environmental conditions of MIL-T-28800.

a. Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be 2,000 hours.

(1) Thermal cycling. Thermal cycling shall expose the equipment to hot and cold cycles for class __ equipment per MIL-T-28800. Each equipment which fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the cycling, a satisfactory operation check (SOC) shall be conducted during the last two upper and lower temperature limits.

10. Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case of modules.

a. Maintainability. The equipment shall meet the type III maintainability requirements specified by MIL-T-28800.

11. Performance characteristics.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, and standards and quality assurance practices and is the same product offered for sale in the commercial

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marketplace. Small business that cannot offer products with established commercial marketplace acceptability, may compete by submission of bid samples. (Government reserves the right to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.)

Packaging, packing, and marking. The items shall be packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with national motor freight classification and uniform freight classification. Shipping container shall be marked as specifiedin the contract or order.

Known acceptable products. The following product(s) are known to have previously met the requirements of this CID and need not to resubmit samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Manufacturer	Model	NSN
Weinschel Engineering	5203	5985-01-212-8733YA

Custodian:
Air Force - 99

Review activity:
Army - CR
DLA - ES

User:
Navy - SH

Preparing activity:
Air Force - 82

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